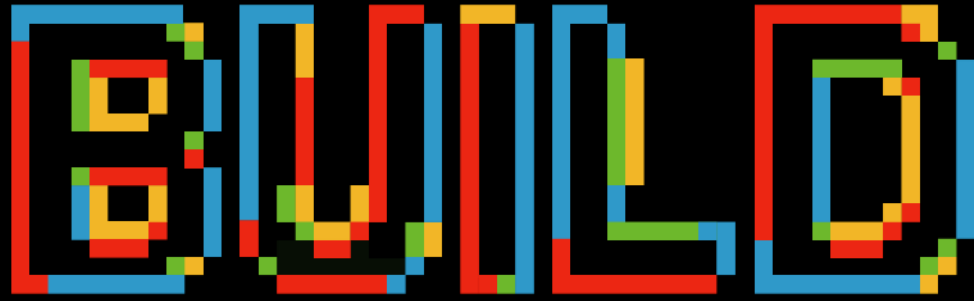




//localhost:Dubai – June 9, 2026

Custom Agent Harnesses with Microsoft Foundry and Agent Framework



Gianni Rosa Gallina
Head of R&D



Trust + Security

Human & Agent Collaboration

Agent Runtime

Intelligence

Simplicity



Control

From chat loop to agent runtime

The hard part is not sending prompts.

The hard part is making agent behavior **controlled, observable, and repeatable.**

The last 6 months in agents

December 2025

June 2026

Model Got
Scary Good
at Coding

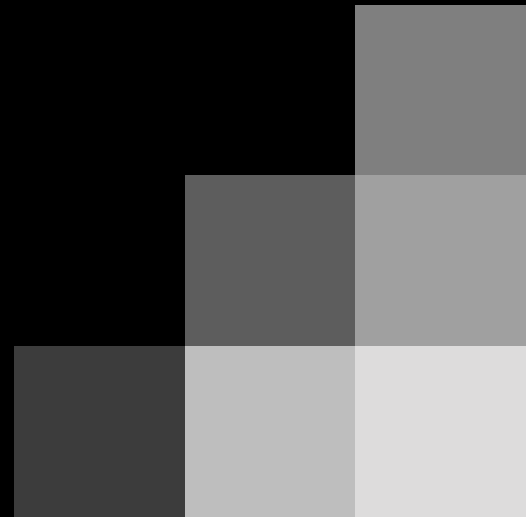
Coding
Agent
Wars

The
"Claw"
Ecosystem

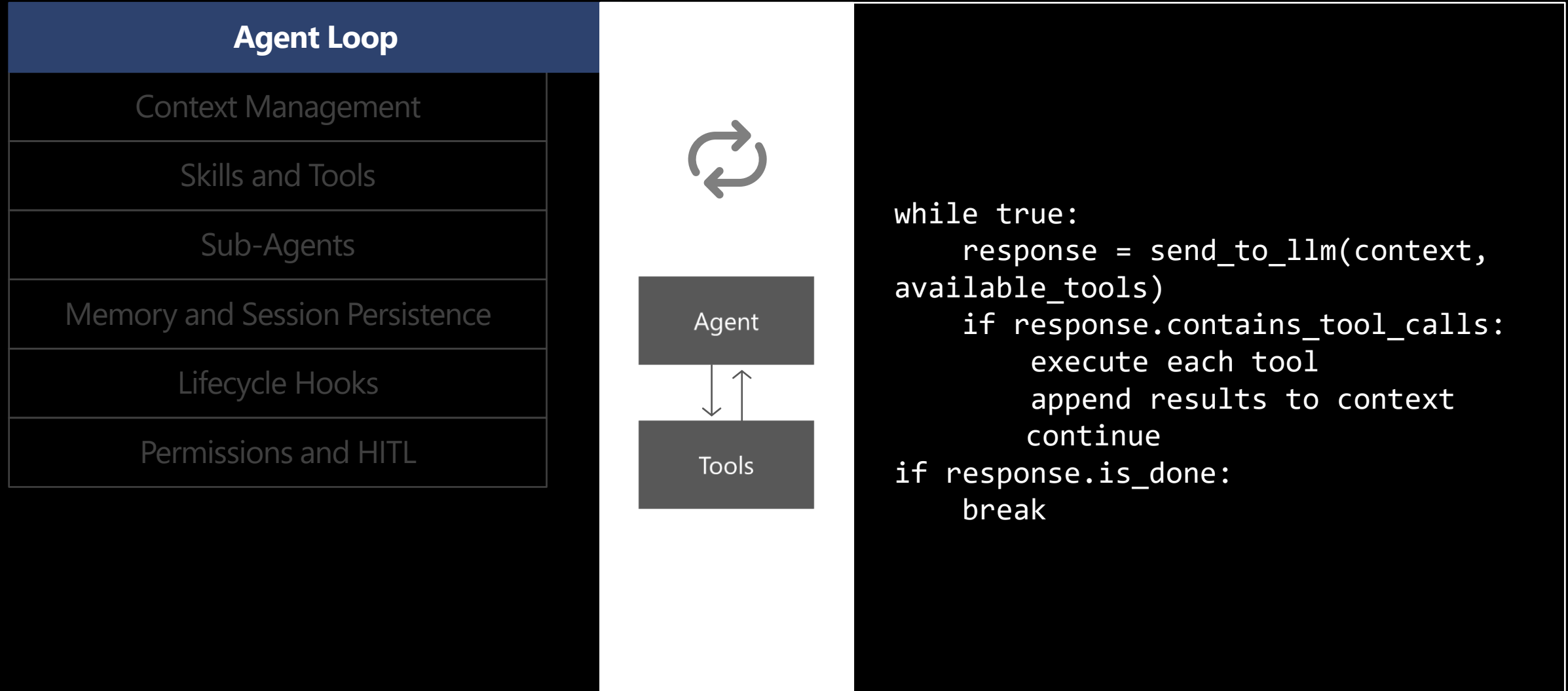
The
Second Wave:
Hermes Agent

Everyone
Gets
a Harness

What is an
Agent Harness?



Components of an agent harness



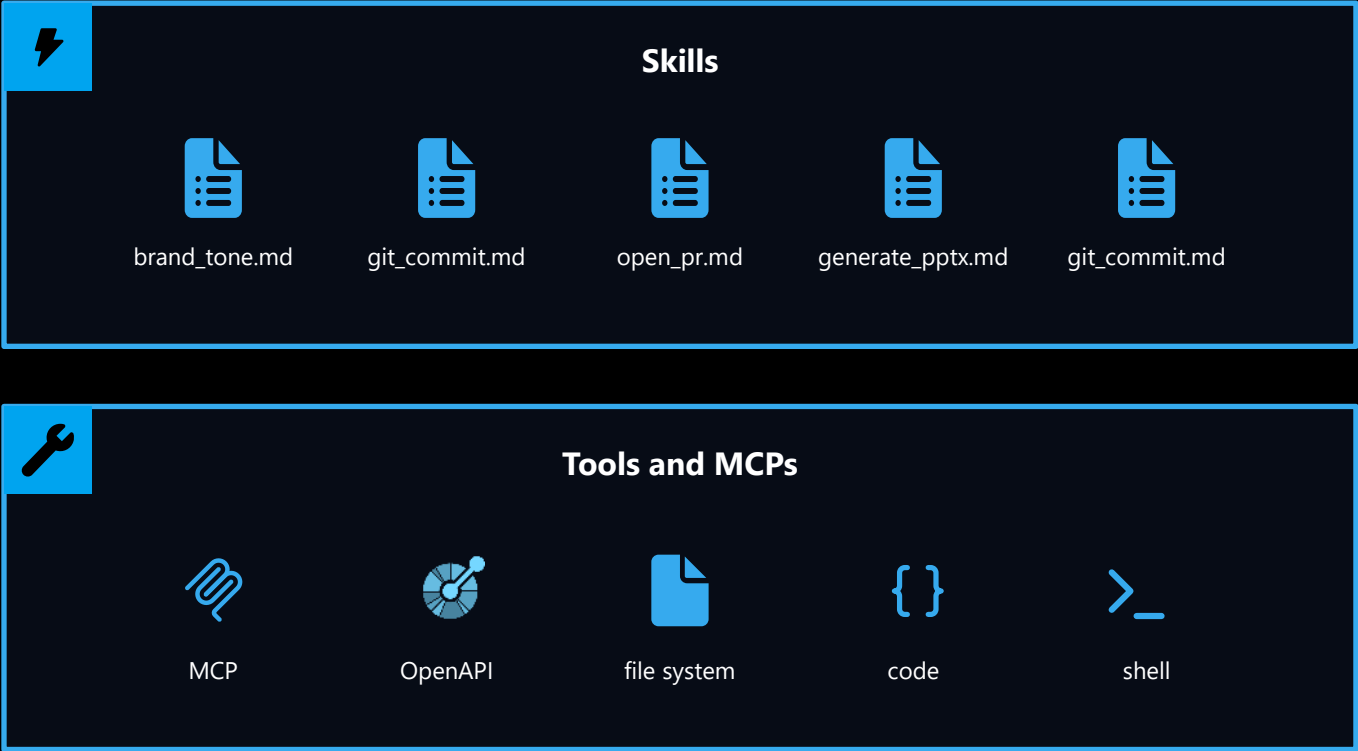
Components of an agent harness

Agent Loop
Context Management
Skills and Tools
Sub-Agents
Memory and Session Persistence
Lifecycle Hooks
Permissions and HITL



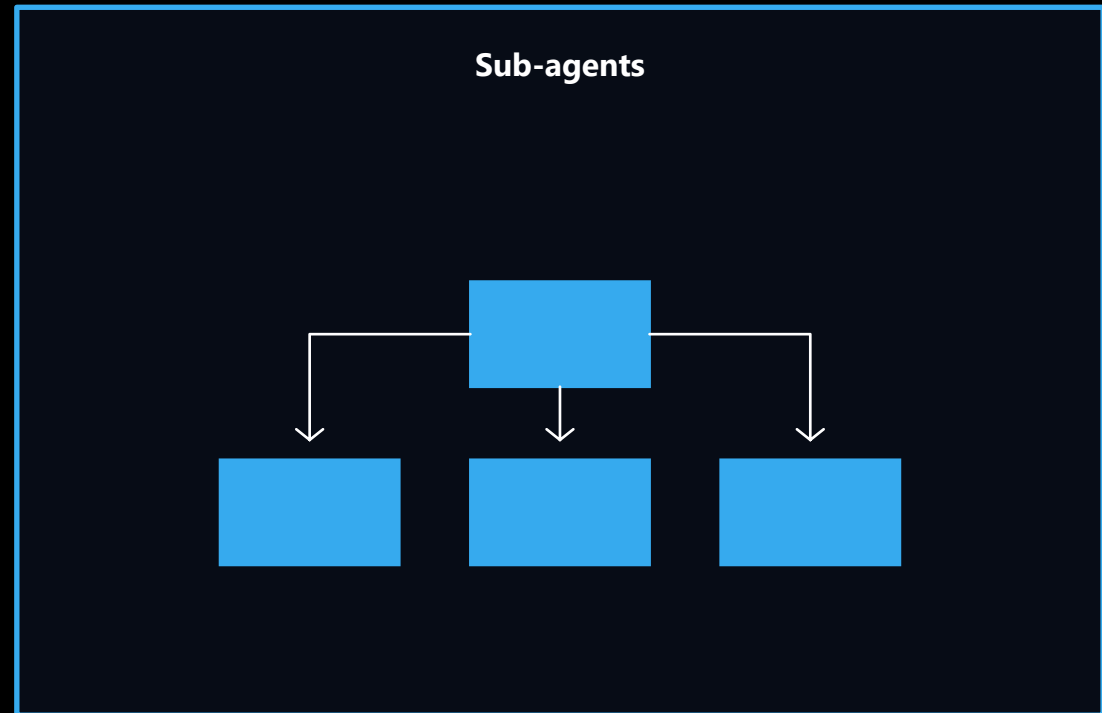
Components of an agent harness

Agent Loop
Context Management
Skills and Tools
Sub-Agents
Memory and Session Persistence
Lifecycle Hooks
Permissions and HITL

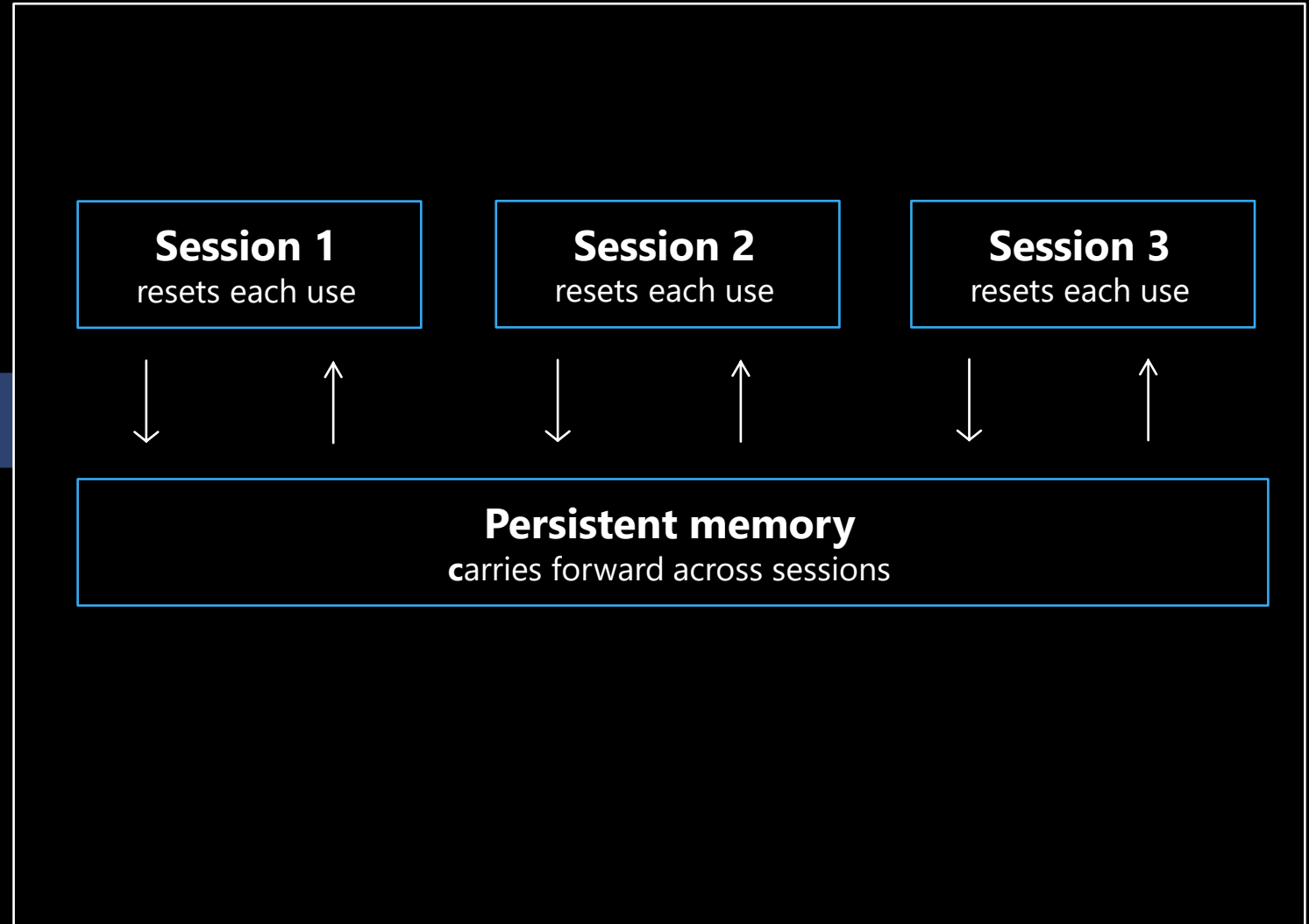
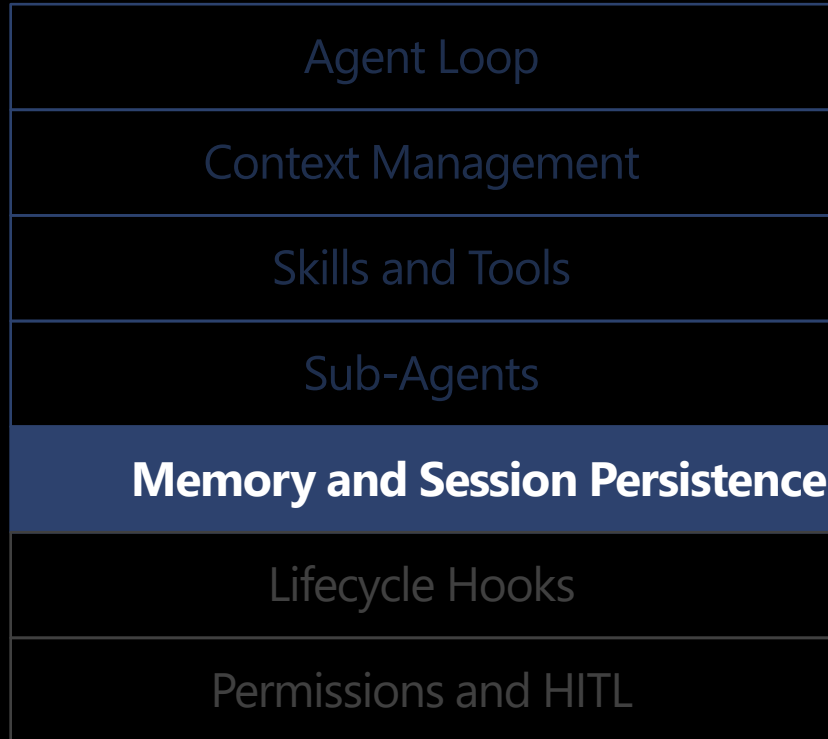


Components of an agent harness

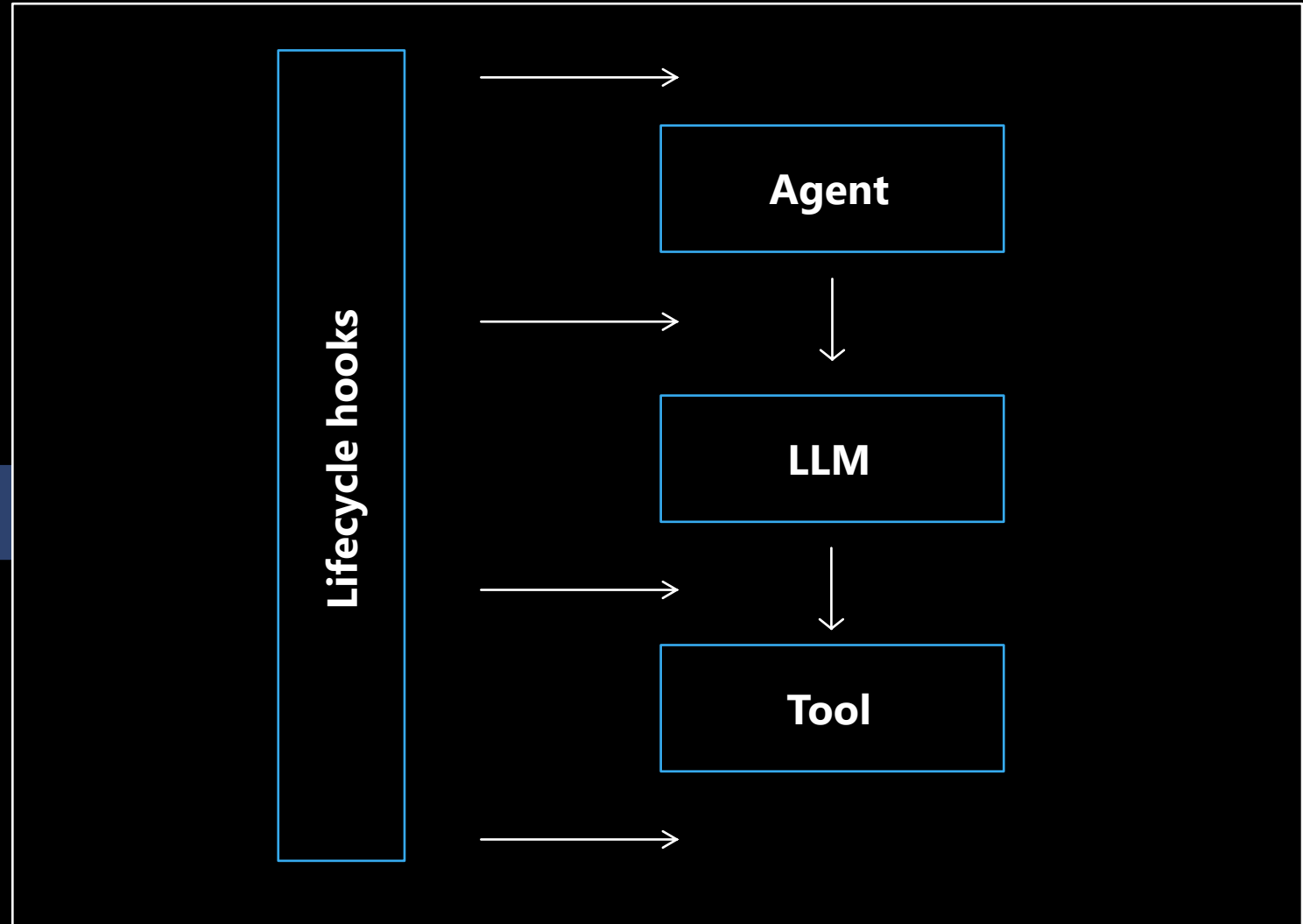
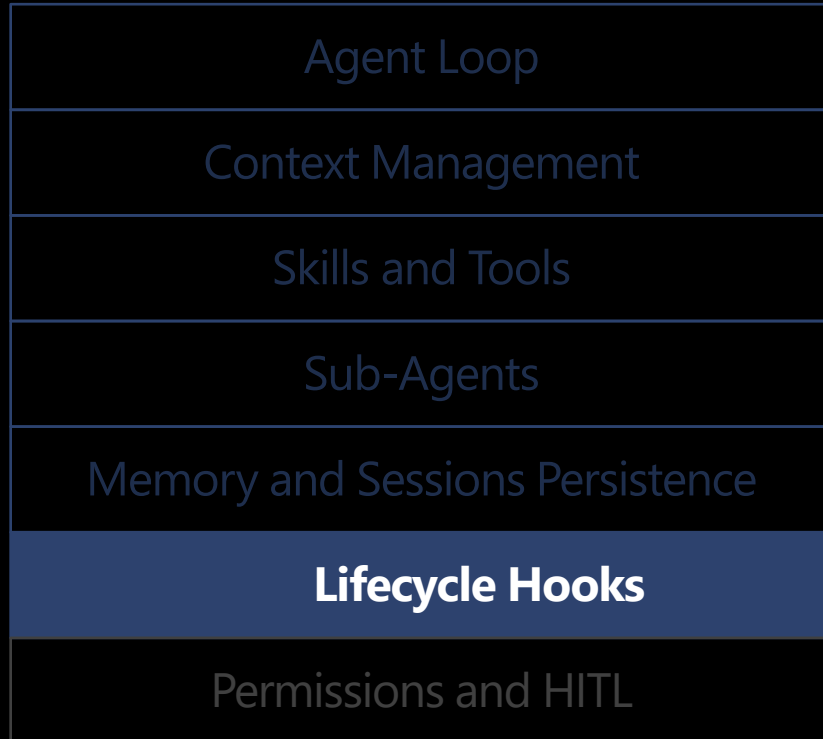
Agent Loop
Context Management
Skills and Tools
Sub-Agents
Memory and Session Persistence
Lifecycle Hooks
Permissions and HITL



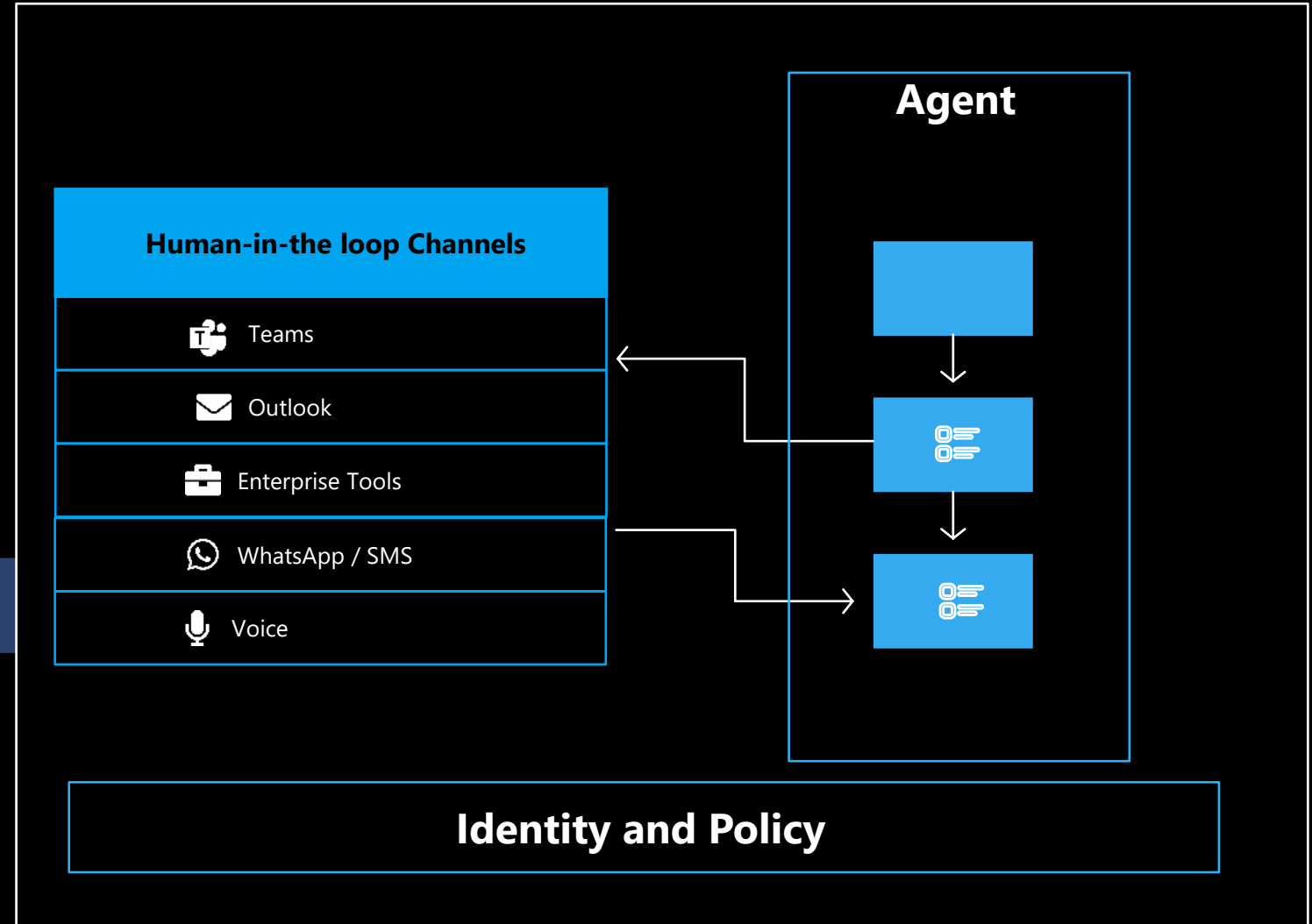
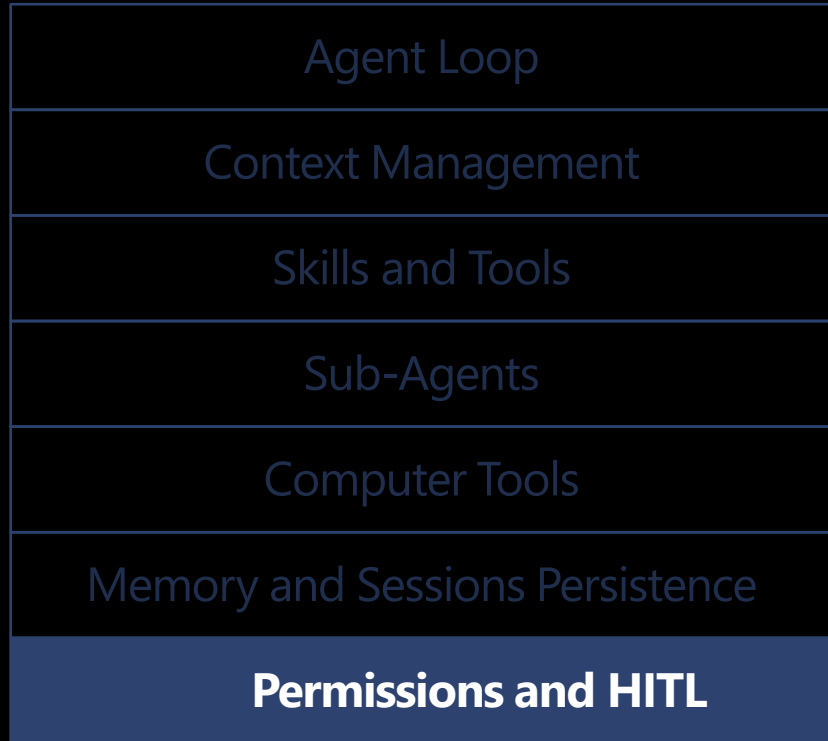
Components of an agent harness



Components of an agent harness



Components of an agent harness





Microsoft Agent Framework

Version 1.0

aka.ms/agent-framework

Workflows

Harness

Agent Loop

Agent Loop



MAF AI Agent

MAF Agent	
 Foundry Models	    
 Foundry Tools	 
 Foundry Hosting	 
Microsoft FoundryEcosystem	

Agent Providers

 Foundry Prompt Agent

 Copilot Studio Agent

 A2A

 Claude Code Agent

 GitHub Copilot CLI Agent

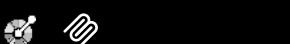
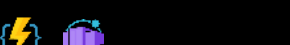
Workflows

Harness

Agent Loop



MAF Agent

 Foundry Models	
 Foundry Tools	
 Foundry Hosting	

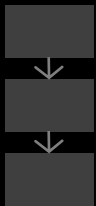
Agent Providers

 Foundry Prompt Agent	 Copilot Studio Agent	 A2A
 Claude Code Agent	 GitHub Copilot CLI Agent	

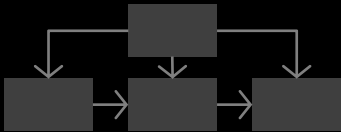
Workflows



Sequential



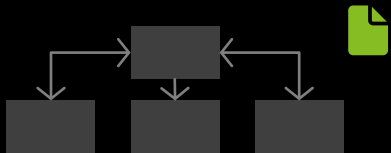
Handoff



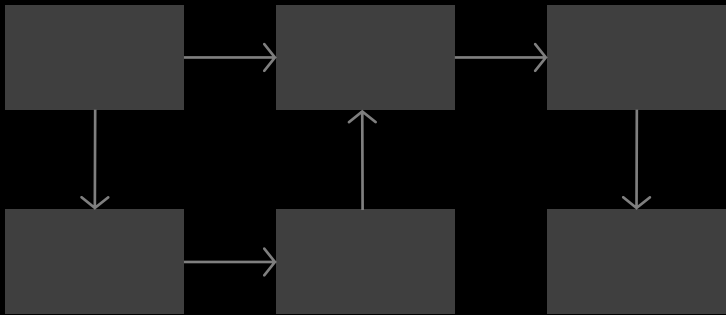
Author/Critic



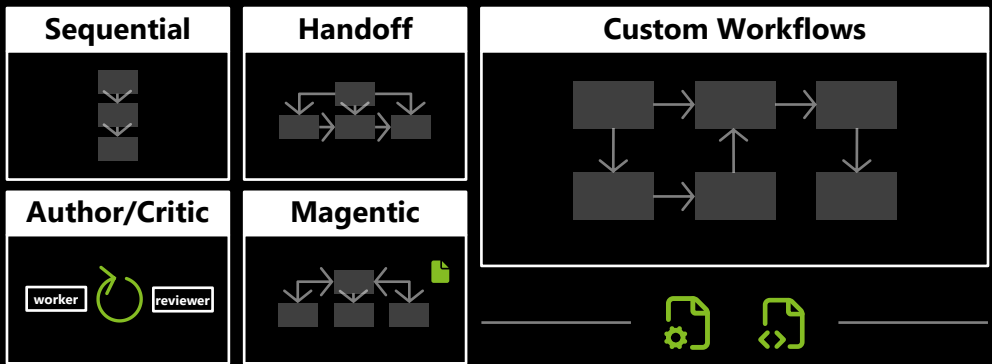
Magnetic



Custom Workflows



Workflows



Harness

Agent Loop



MAF Agent	
Foundry Models	AI
Foundry Tools	
Foundry Hosting	

Agent Providers



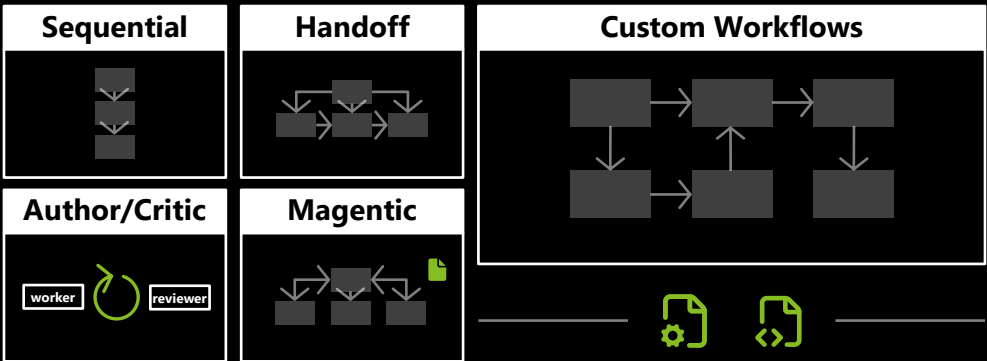
Harness



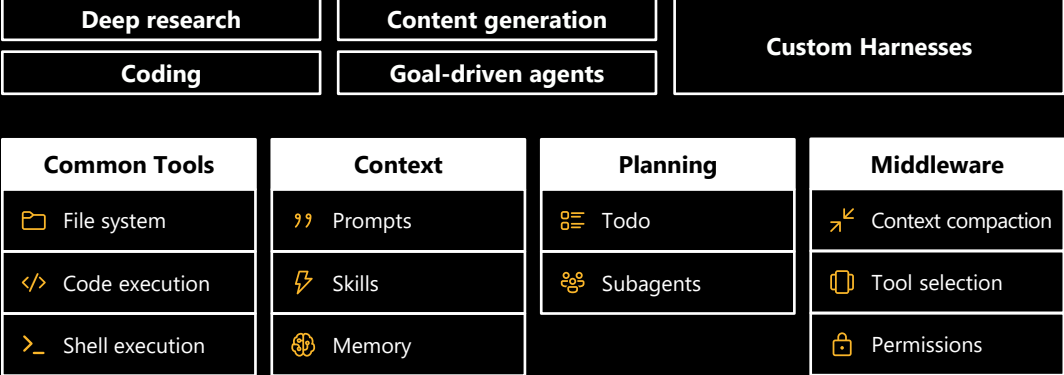
Deep research	Content generation	Custom Harnesses
Coding	Goal-driven agents	

Common Tools	Context	Planning	Middleware
 File system	 Prompts	 Todo	 Context compaction
 Code execution	 Skills	 Subagents	 Tool selection
 Shell execution	 Memory		 Permissions

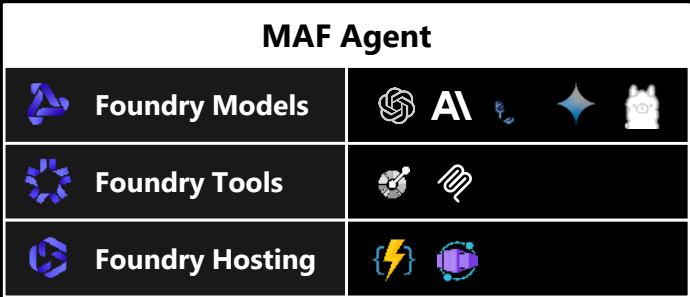
Workflows



Harness



Agent Loop



Agent Providers



Agent Harness in Agent Framework

#	Component	BUILD	Package
1	Agent loop	Agent Harness Loop in Public Preview	Preview 
2	Context Management	Context Management at v1.0	Release 
3	Skills and tools	Skills at v1.0	Release 
4	Sub-agents	Auto sub-agents in preview	Preview 
5	Session persistence (memory, survives crash)	Workflows at v1.0	Release 
6	System prompt assembly	File-based memory pipeline	Preview 
7	Lifecycle hooks	Agent middleware	Release 
8	Permissions and safety	Tools permission in preview	Preview 
9	HIL and planning	Preview with Harness	Preview 

Microsoft Agent Framework New Features

#	Component	BUILD	Package
1	Copilot SDK Integration		Release
2	Claude Code SDK Integration		Preview
3	Orchestrations Workflows Patterns		Release
4	CodeAct AIProvider		Preview

Demo

Custom Agents and Harnesses with Microsoft Agent Framework & Foundry

<https://github.com/gianni-rg/Build26-Localhost-Dubai-custom-agent-harness>



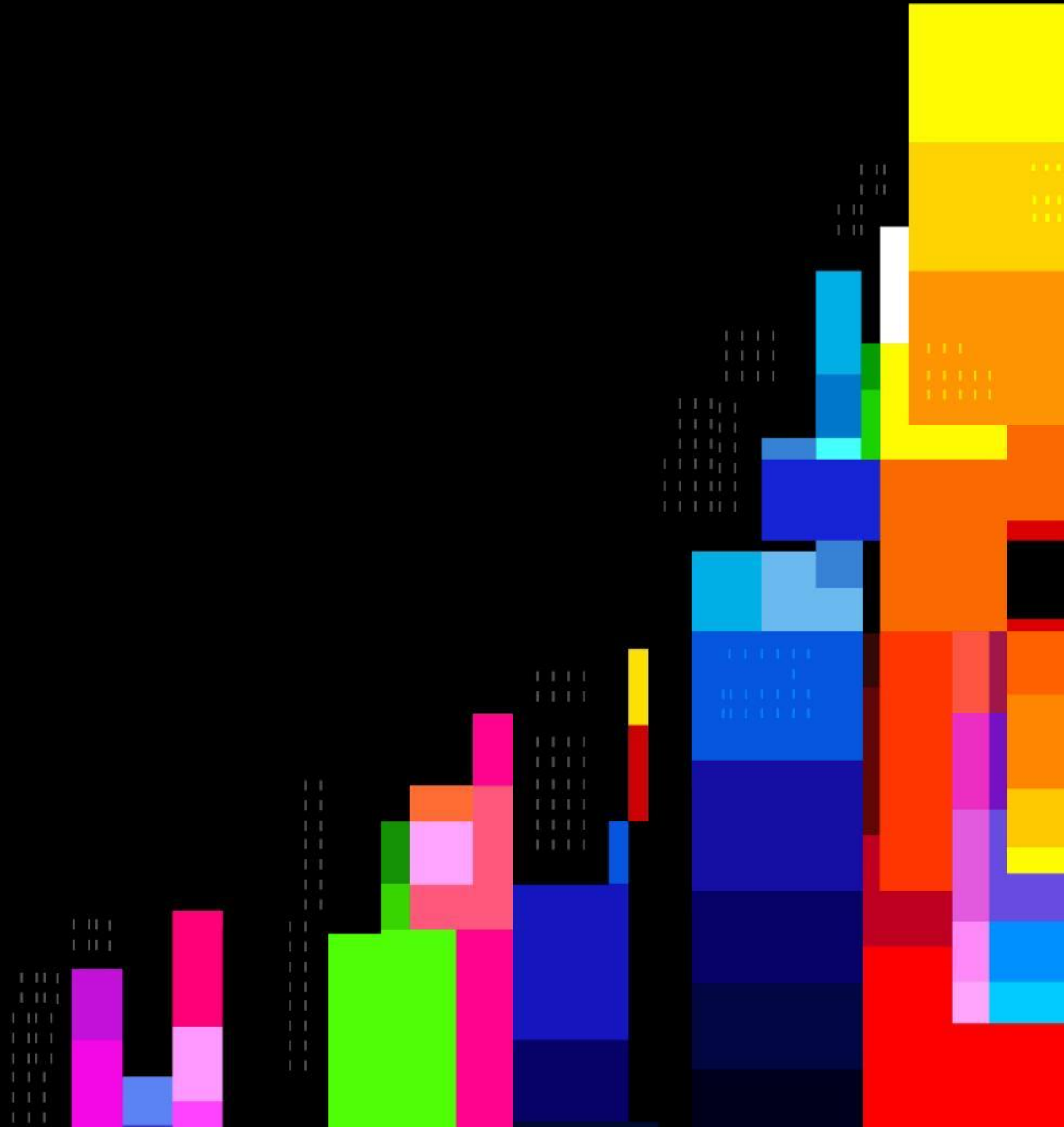
Demo – Step 1

Planning, modes, todos, approval



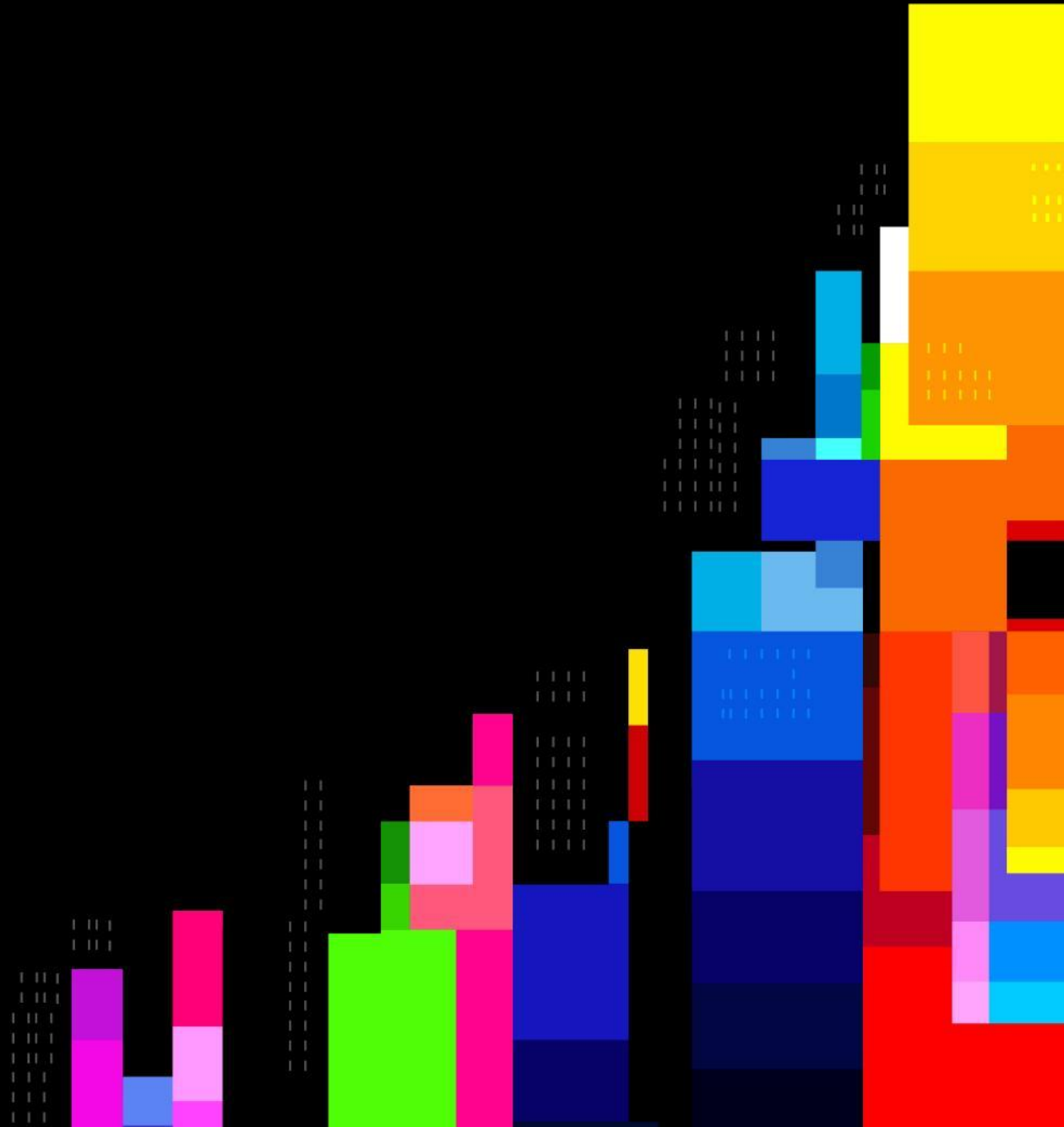
Demo – Step 2

Background agents



Demo – Step 3

File-grounded work

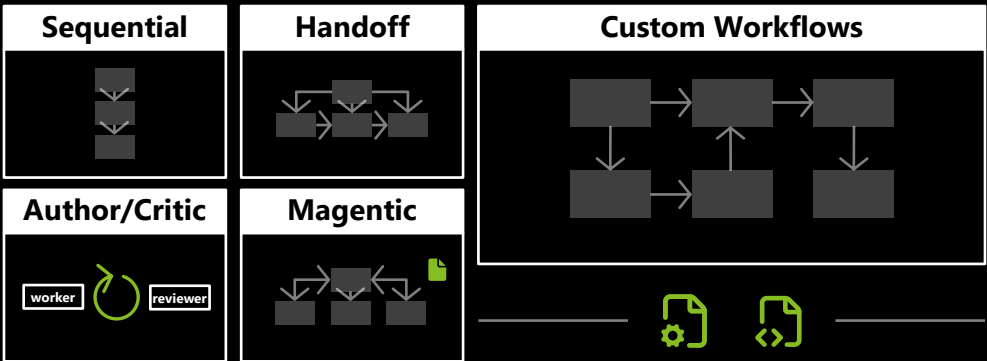


Demo – Step 4

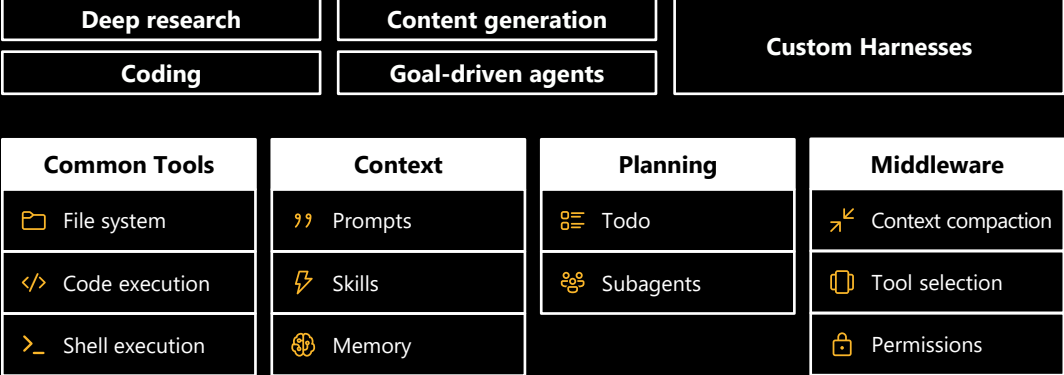
Skills and sandboxed code execution



Workflows



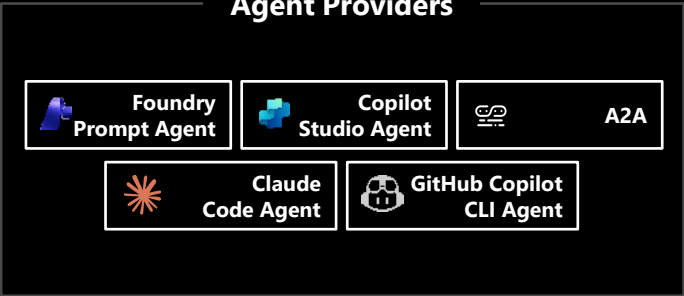
Harness



Agent Loop



MAF Agent	
Foundry Models	AI
Foundry Tools	
Foundry Hosting	



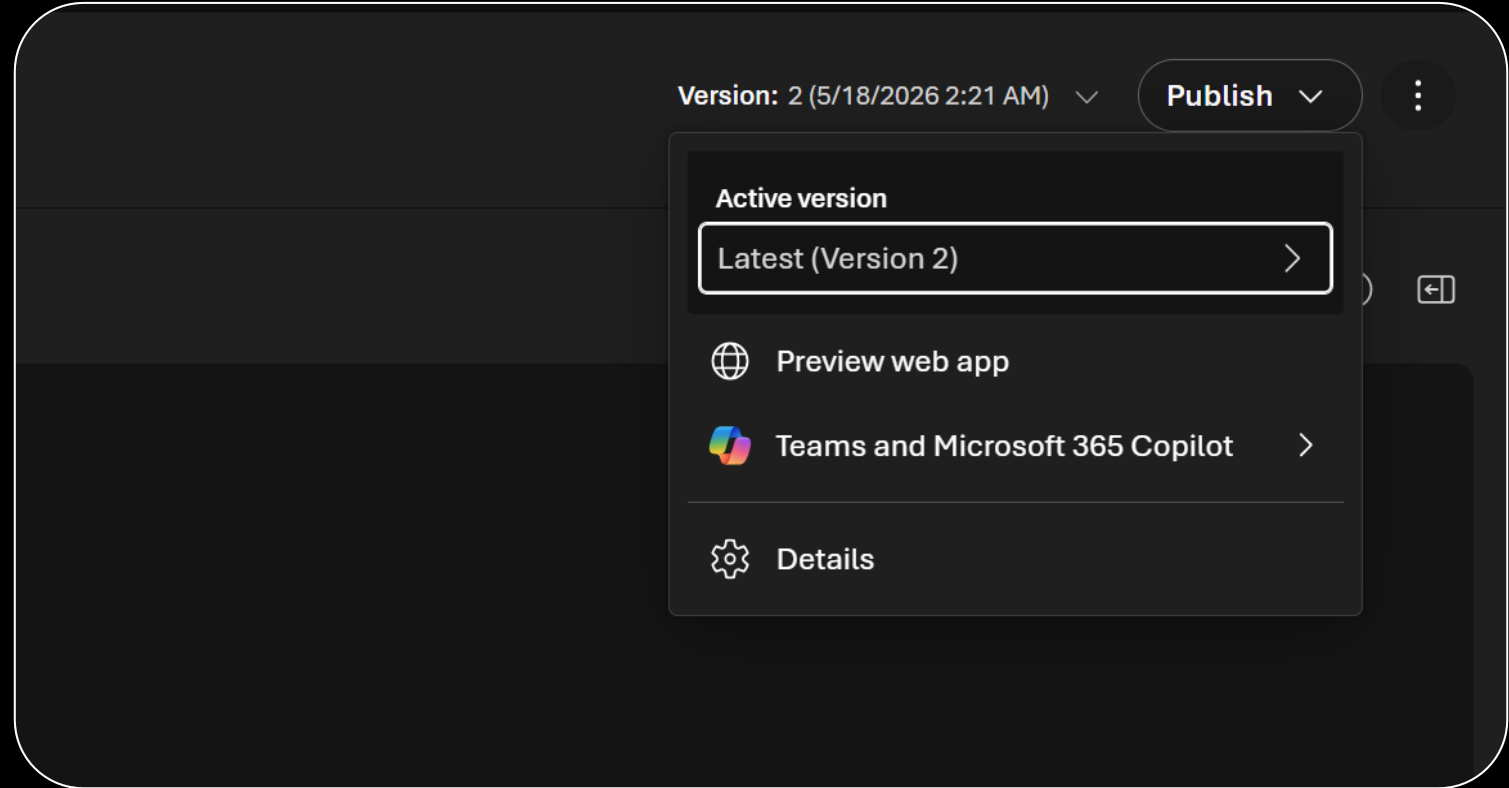
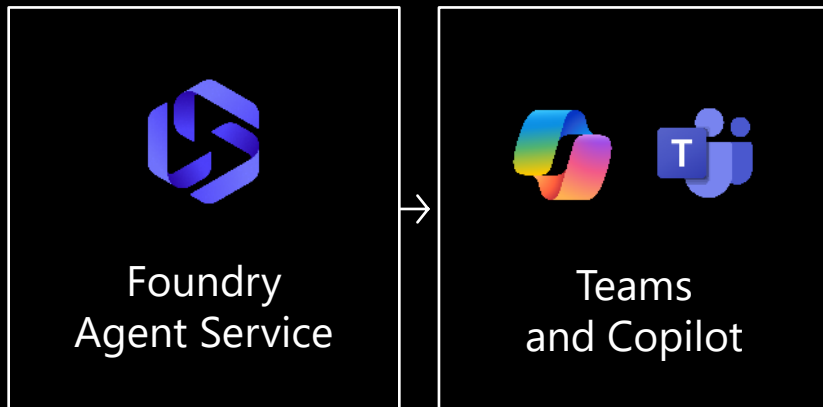
Microsoft Agent Platform

Build in GitHub. Run and optimize in Foundry. Reach users in M365, Teams, and everywhere work gets done.



Publish to Teams and Microsoft 365 Copilot

Take agents built with Foundry and make them available instantly in M365 Copilot and Teams Chat



With you, every step of the way



Start building with Microsoft Foundry

ai.azure.com



Get the code from this session

github.com/gianni-rg/Build26-Localhost-Dubai-custom-agent-harness



Join Foundry Discord community

aka.ms/foundry/discord

Agents League Hackathon



Creative
Apps



Reasoning
Agents



Enterprise
Agents

Learn more



May 19th – June 14th

Register now, submit your
project and compete for prizes!

aka.ms/agentsleague/aisf

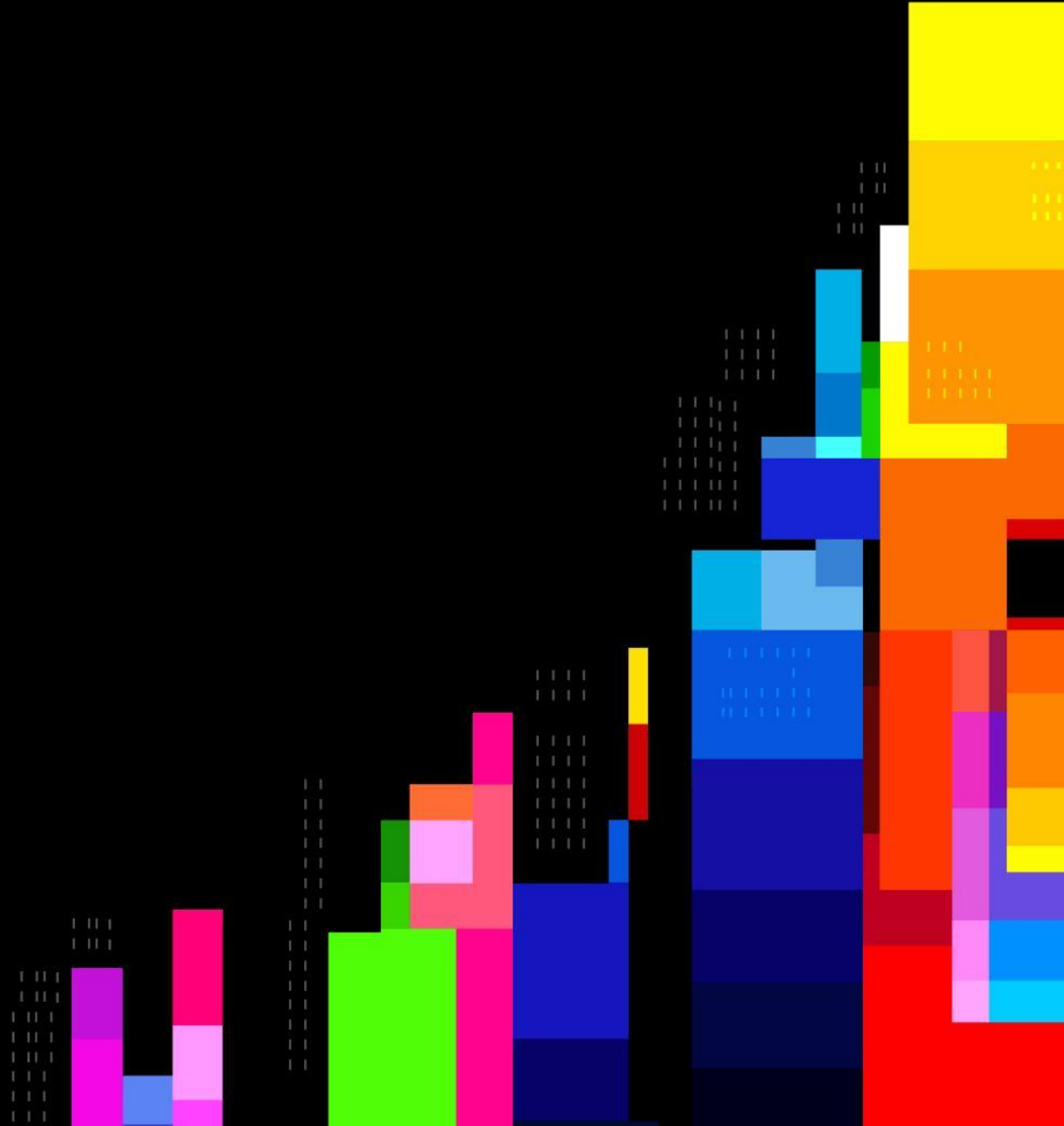
Thank you!

Let's keep the conversation going – stop by to get your questions answered.

Session code	Title	Date & time
BRK240	Build context-aware agents: From data to decisions	Tuesday 6/2 5:00 PM
BRK252	From Observability to ROI for AI agents on any framework	Tuesday 6/2 5:00 PM
BRK246	Foundry IQ: Fuel agents with enterprise knowledge and agentic retrieval	Wednesday 6/3 9:00 AM
BRK243	Claw and agent harness in Microsoft Foundry	Wednesday 6/3 11:30 AM
BRK244	Agent supervision is the senior engineering skill	Wednesday 6/3 10:15 AM
BRK242	Turn your agents into action: Connect tools, APIs, and documents	Wednesday 6/3 4:00 PM
DEM333	How Foundry integrates with open-source frameworks and tools	Wednesday 6/3 4:00 PM

build.microsoft.com

Questions?



About me



Microsoft
Specialist

Programming in C#
Programming in HTML5
with JavaScript & CSS3

Microsoft
CERTIFIED

Solutions Developer

Windows Store Apps Using C#
Web Applications



PLURALSIGHT
Author

Ing. Gianni **ROSA GALLINA** Head of R&D

X@giannirg

- Generative AI, Agents, Agentic Engineering & Harnesses
- Machine Learning, Deep Learning on multi-modal content
- Virtual/Augmented/Mixed Reality
- Immersive video streaming & 3D graphics
- Cloud solutions, web backends, serverless, video workflows
- Mobile apps dev (Windows / Android / .NET MAUI / Avalonia)
- End-to-end solutions with Microsoft Azure
- Blog: <https://gianni.rosagallina.com/en/>
- LinkedIn: <https://www.linkedin.com/in/gianni-rosa-gallina/>
- GitHub: <https://github.com/gianni-rg>

